AA256 HW4 Solution

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1 Thermal response

Both laminates have four 0.007 in plies (t = 0.028 in). Use z positive downward, with top face z = −0.014 in. ΔT = 75 − 400 = −325°F. Global engineering strain order is (ε_1, ε_2, γ_{12}).

$$\{\varepsilon^T\}local = \begin{bmatrix} 0.002210 \\ -0.010400 \\ 0 \end{bmatrix}$$

For a 90° ply, swap the two normal thermal strains. Integrate the transformed ply stiffness $\bar{Q}$ and thermal strain through the thickness:

$$N^T = \sum Q_k \varepsilon^T{}_k \, (z_k - z_{k-1})$$

$$M^T = \frac{1}{2} \sum Q_k \varepsilon^T{}_k \, ({z_k}^2 - {z_{k-1}}^2)$$

$$\begin{bmatrix} A & B \\ B & D \end{bmatrix} \begin{bmatrix} \varepsilon^0 \\ \kappa \end{bmatrix} = \begin{bmatrix} N^T \\ M^T \end{bmatrix}$$

N^T and M^T are equivalent thermal loads. The actual net force and moment resultants are zero because cooling is mechanically unrestrained.

Quantity	[0/90/90/0]	[0/0/90/90]
N^T_1 (lbf/in)	488.93972	488.93972
N^T_2 (lbf/in)	488.93972	488.93972
M^T1 (lbf)	0	−5.92582
M^T2 (lbf)	0	5.92582
$\varepsilon^o1 \times 10^3$	1.37826	−0.67598
$\varepsilon^o2 \times 10^3$	1.37826	−0.67598
$\kappa1$ (in^{-1})	0	−0.33806
$\kappa2$ (in^{-1})	0	0.33806

All shear/twist components are zero. The symmetric laminate has B = 0 and remains flat. The unsymmetric laminate develops opposite-sign normal curvatures. Reversing the z convention reverses B, M^T and κ consistently.

Residual fields for [0/90/90/0]

For each ply, evaluate both faces using $\varepsilon(z) = \varepsilon^0 + z\kappa$, $\varepsilon^R(z) = \varepsilon(z) - \varepsilon^T$, and $\sigma^R(z) = \bar{Q}\,\varepsilon^R(z)$. The tables use global axes; shear strain and shear stress are zero throughout. T and B denote the top and bottom faces of each ply.

Ply	Face	z (in)	$\varepsilon_1 \times 10^3$	$\varepsilon_2 \times 10^3$	$\varepsilon^R_1 \times 10^3$	$\varepsilon^R_2 \times 10^3$
1	T	-0.014	1.3783	1.3783	−0.8317	11.7783
1	B	-0.007	1.3783	1.3783	−0.8317	11.7783
2	T	-0.007	1.3783	1.3783	11.7783	−0.8317
2	B	0.000	1.3783	1.3783	11.7783	−0.8317
3	T	0.000	1.3783	1.3783	11.7783	−0.8317
3	B	0.007	1.3783	1.3783	11.7783	−0.8317
4	T	0.007	1.3783	1.3783	−0.8317	11.7783
4	B	0.014	1.3783	1.3783	−0.8317	11.7783

Residual stresses

Ply	Angle	Face	σ_1 (ksi)	σ_2 (ksi)
1	0	T	−15.0752	15.0752
1	0	B	−15.0752	15.0752
2	90	T	15.0752	−15.0752
2	90	B	15.0752	−15.0752
3	90	T	15.0752	−15.0752
3	90	B	15.0752	−15.0752
4	0	T	−15.0752	15.0752
4	0	B	−15.0752	15.0752

Total strain is continuous through interfaces; elastic strain and stress can jump. For local ply results, interchange normal components in the 90° plies. Positive stress denotes tension. Integrating stresses gives zero net force and moment.

Residual fields for [0/0/90/90]

For each ply, evaluate both faces using $\varepsilon(z) = \varepsilon^0 + z\kappa$, $\varepsilon^R(z) = \varepsilon(z) - \varepsilon^T$, and $\sigma^R(z) = \bar{Q}\,\varepsilon^R(z)$. The tables use global axes; shear strain and shear stress are zero throughout. T and B denote the top and bottom faces of each ply.

Ply	Face	z (in)	$\varepsilon_1 \times 10^3$	$\varepsilon_2 \times 10^3$	$\varepsilon^R_1 \times 10^3$	$\varepsilon^R_2 \times 10^3$
1	T	-0.014	4.0569	−5.4088	1.8469	4.9912
1	B	-0.007	1.6904	−3.0424	−0.5196	7.3576
2	T	-0.007	1.6904	−3.0424	−0.5196	7.3576
2	B	0.000	−0.6760	−0.6760	−2.8860	9.7240
3	T	0.000	−0.6760	−0.6760	9.7240	−2.8860
3	B	0.007	−3.0424	1.6904	7.3576	−0.5196
4	T	0.007	−3.0424	1.6904	7.3576	−0.5196
4	B	0.014	−5.4088	4.0569	4.9912	1.8469

Residual stresses

Ply	Angle	Face	σ_1 (ksi)	σ_2 (ksi)
1	0	T	44.8609	7.1924
1	0	B	−9.4171	9.4171
2	0	T	−9.4171	9.4171
2	0	B	−63.6951	11.6419
3	90	T	11.6419	−63.6951
3	90	B	9.4171	−9.4171
4	90	T	9.4171	−9.4171
4	90	B	7.1924	44.8609

Total strain is continuous through interfaces; elastic strain and stress can jump. For local ply results, interchange normal components in the 90° plies. Positive stress denotes tension. Integrating stresses gives zero net force and moment.

2 Lab coupon comparison

Use the supplied T800SC/3900-2B coupon properties: E_{11} = 21.5 msi, E_{22} = 1.20 msi, ν_{12} = 0.018 and G_{12} = 0.570 msi. The table below uses the supplied measured width and thickness; all four plies are assumed equally thick. Form the full ABD matrix for each layup and let α be the upper-left block of its inverse.

$$E_1 = \frac{1}{t\,\alpha_{11}}, \quad P = \frac{E_1\, b\, t}{Lg} u$$

Here u is extension over Lg. This definition uses midplane strain, with no applied moments and free bending. Curvature is already included through $\alpha = (A - BD^{-1}B)^{-1}$. Using A^{-1} would instead suppress curvature.

Coupon	E_1 (msi)	E_1 (GPa)	Slope* (N/mm)
1	21.50000	148.2373	12085.70
2	11.35016	78.2566	5257.77
3	4.54239	31.3187	2083.67
4	6.58943	45.4325	3742.37
5	1.82328	12.5711	994.39
6	2.07226	14.2877	1162.95
7	11.40429	78.6298	6444.56
8	2.99871	20.6754	1667.22
9	12.98083	89.4997	7420.72
10	4.31891	29.7779	2463.74

*Reference slope uses the listed specimen length as Lg. These lengths are not confirmed displacement gauge lengths. For a measured gauge length, replace the reference slope by Knew = Ktable Llisted/Lg, using consistent length units.

Overlay each measured load–extension curve with the corresponding straight elastic line, using the actual gauge length. If fitting an experimental modulus, fit the initial linear loading region with an intercept: Eexp = (dP/du)Lg/(bt). Exclude seating, slip, unloading and post-failure records. Crosshead travel also includes machine and grip compliance.

Expected discussion: unsymmetric stacks can bend or twist; grips may restrain those motions. Thickness, angle, material scatter, displacement definition, and the selected linear interval also affect measured slopes. The line is an elastic prediction, not a failure-load prediction.

The following pages contain the measured comparisons and fitted slopes from all eleven supplied test records. Coupon 1 has separate interrupted and restarted runs. The MATLAB script reproduces these plots and tables.

2.1 Experimental reduction and results

Read force from column 3 (N) and axial machine displacement from column 4 (mm), skipping repeated MTS header blocks. The recorded strain channel is nearly constant and is not used. Subtract the median of the first 20 samples from force and displacement separately for each run.

Fit force versus displacement with an intercept, using points between 10% and 40% of that run's peak force, before the peak. This automatic interval is a reproducible initial-slope estimate; review each zoom plot for seating or nonlinearity. The measured slope Kexp does not require a gauge-length assumption.

$$Kexp = \frac{dP}{du}, \quad Eapp = \frac{Kexp\ Lg}{bt}$$

Run	Ktheory* (N/mm)	Kexp (N/mm)	Eapp* (msi)	Fit R^2
1 initial	12085.7	9918.6	17.6447	0.99995
1 restart	12085.7	9522.9	16.9409	0.99992
2	5257.8	4416.4	9.5339	0.99990
3	2083.7	4443.7	9.6872	0.99994
4	3742.4	2786.7	4.9067	0.99907
5	994.4	1142.9	2.0955	0.99808
6	1163.0	1136.9	2.0259	0.99776
7	6444.6	5688.7	10.0666	0.99992
8	1667.2	3104.9	5.5846	0.99942
9	7420.7	7759.4	13.5732	0.99996
10	2463.7	2913.2	5.1069	0.99925

*Ktheory and Eapp use the listed specimen lengths as Lg. Actual grip-to-grip lengths are not specified in the supplied export. Update those lengths before interpreting these as quantitative material-modulus comparisons. Machine displacement also includes grip and machine compliance.

Coupon 1's first test stopped before failure; the second restarted from zero load. Keep the runs separate. A high fit R^2 measures linearity within the selected interval, not the accuracy of the assumed length or boundary conditions.

Coupons 3 and 8 are substantially stiffer experimentally than the free-bending prediction. Grip restraint of extension–bending coupling is a possible contributor, but these curves alone do not establish the cause. Geometry, actual measurement length, material variability and the supplied $\nu_{12} = 0.018$ should also be checked. Coupon 6 shows a long nonlinear response; only its initial slope is compared to linear theory.

2.2 Coupons 1 and 2

Coupon 1

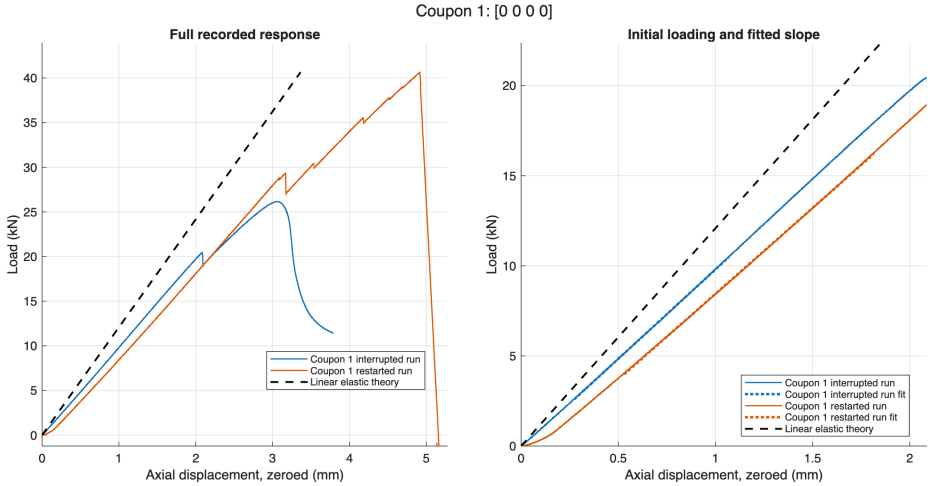


Figure 1. Full recorded response (left) and initial-loading detail (right). Solid curves are measured data, the dashed black line is elastic theory, and dotted curves are the fitted initial response.

Coupon 2

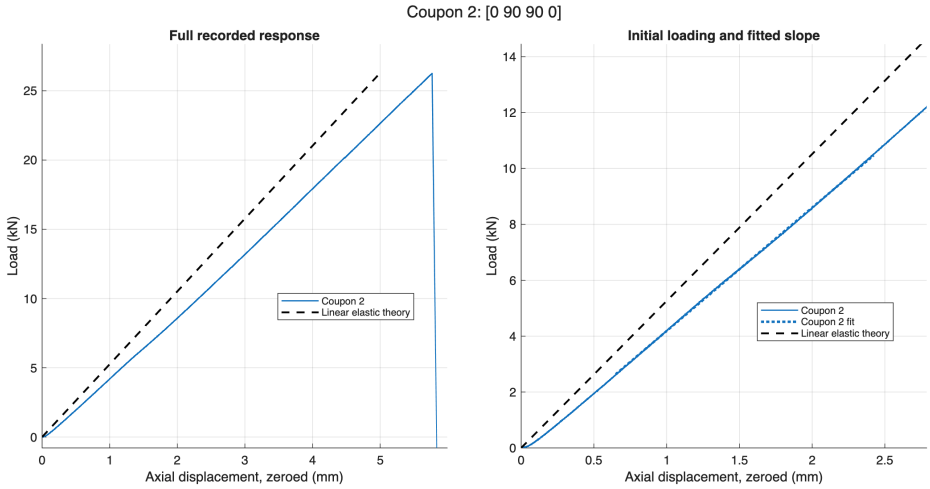


Figure 2. Full recorded response (left) and initial-loading detail (right). Solid curves are measured data, the dashed black line is elastic theory, and dotted curves are the fitted initial response.

Theory uses zero applied moments and free bending, with the listed specimen length as the reference measurement length. Its plotted endpoint is limited by the recorded test range for readability and is not a predicted failure load.

Both coupon 1 runs are plotted separately; neither displacement history is appended to the other.

2.3 Coupons 3 and 4

Coupon 3

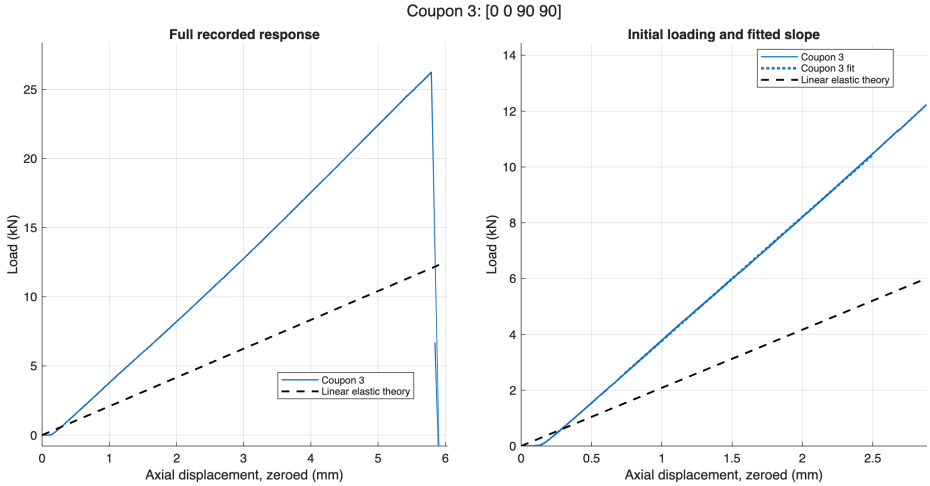


Figure 3. Full recorded response (left) and initial-loading detail (right). Solid curves are measured data, the dashed black line is elastic theory, and dotted curves are the fitted initial response.

Coupon 4

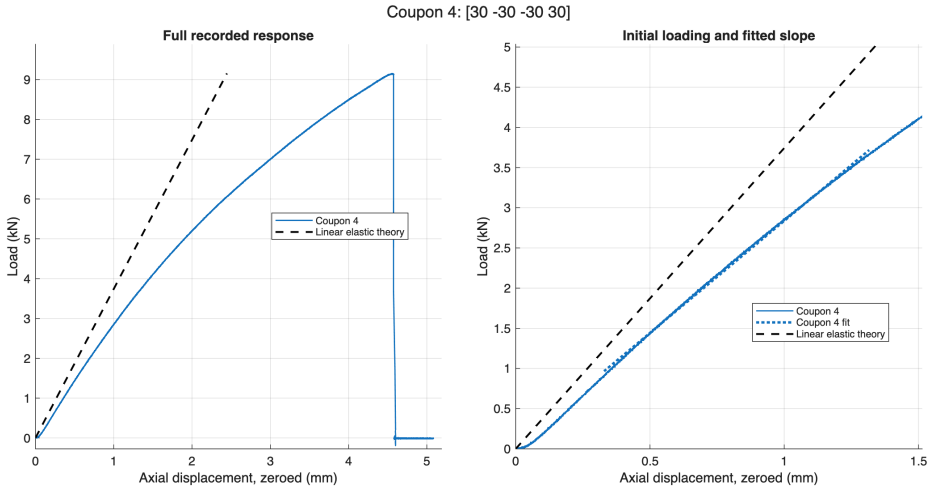


Figure 4. Full recorded response (left) and initial-loading detail (right). Solid curves are measured data, the dashed black line is elastic theory, and dotted curves are the fitted initial response.

Theory uses zero applied moments and free bending, with the listed specimen length as the reference measurement length. Its plotted endpoint is limited by the recorded test range for readability and is not a predicted failure load.

2.4 Coupons 5 and 6

Coupon 5

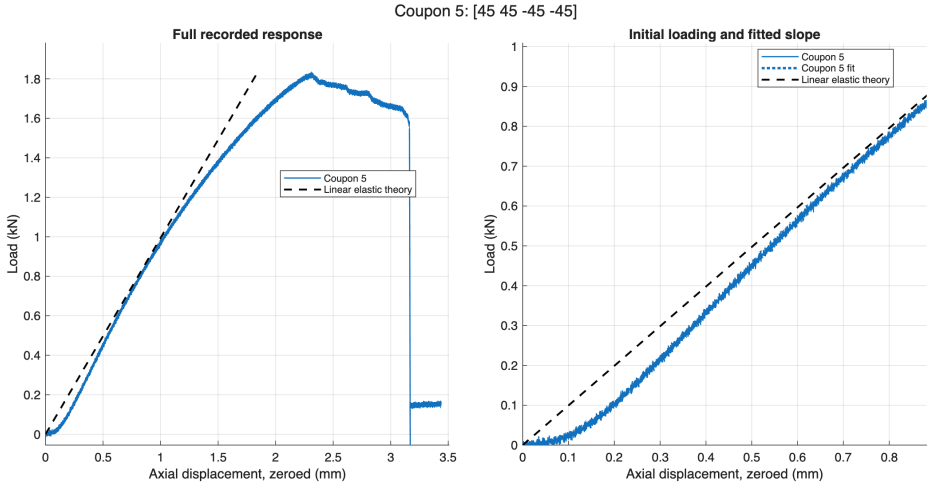


Figure 5. Full recorded response (left) and initial-loading detail (right). Solid curves are measured data, the dashed black line is elastic theory, and dotted curves are the fitted initial response.

Coupon 6

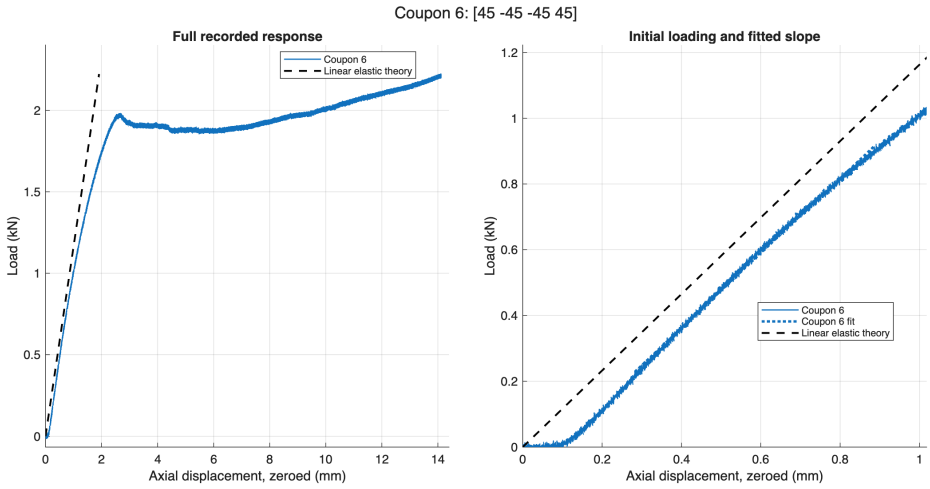


Figure 6. Full recorded response (left) and initial-loading detail (right). Solid curves are measured data, the dashed black line is elastic theory, and dotted curves are the fitted initial response.

Theory uses zero applied moments and free bending, with the listed specimen length as the reference measurement length. Its plotted endpoint is limited by the recorded test range for readability and is not a predicted failure load.

2.5 Coupons 7 and 8

Coupon 7

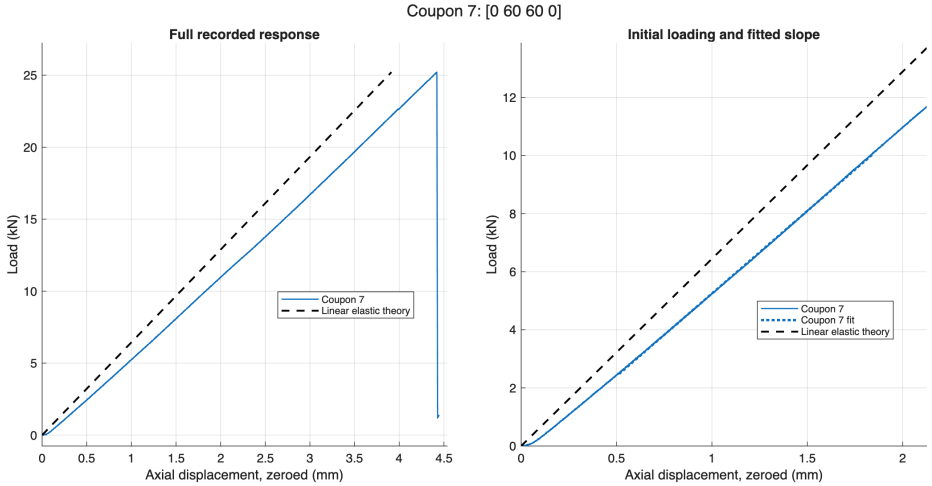


Figure 7. Full recorded response (left) and initial-loading detail (right). Solid curves are measured data, the dashed black line is elastic theory, and dotted curves are the fitted initial response.

Coupon 8

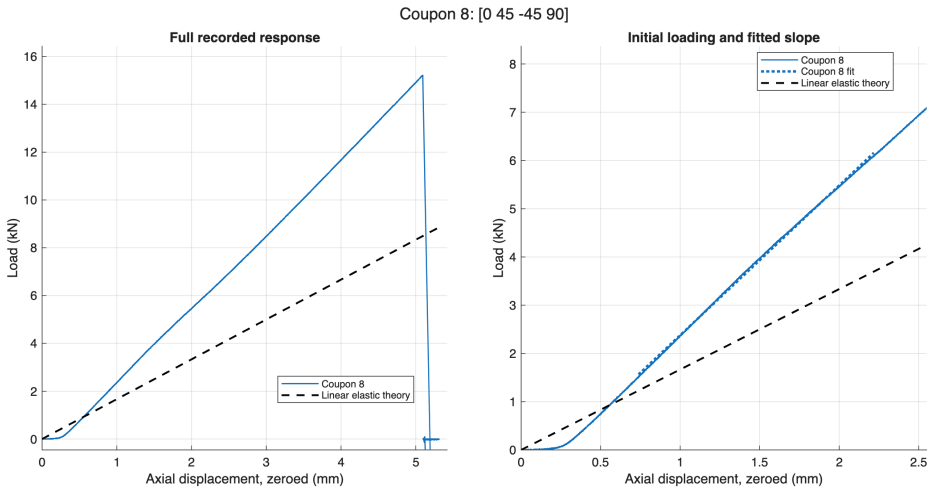


Figure 8. Full recorded response (left) and initial-loading detail (right). Solid curves are measured data, the dashed black line is elastic theory, and dotted curves are the fitted initial response.

Theory uses zero applied moments and free bending, with the listed specimen length as the reference measurement length. Its plotted endpoint is limited by the recorded test range for readability and is not a predicted failure load.

2.6 Coupons 9 and 10

Coupon 9

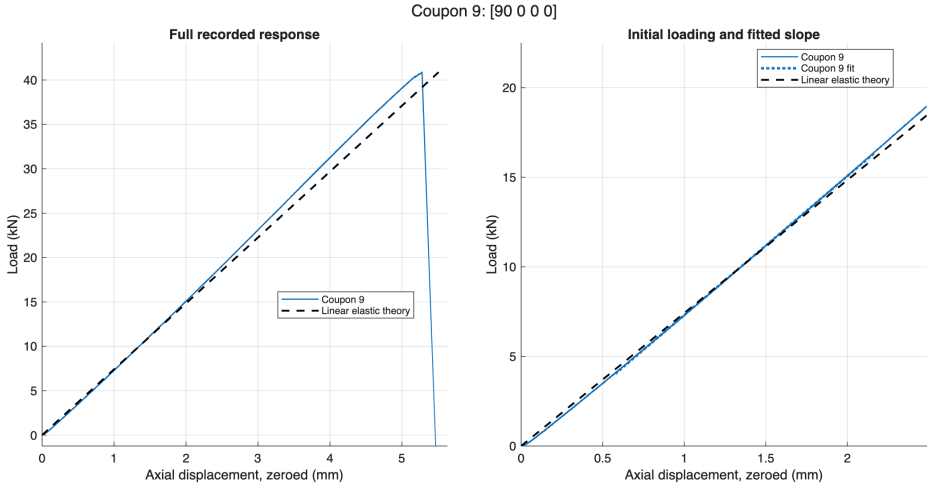


Figure 9. Full recorded response (left) and initial-loading detail (right). Solid curves are measured data, the dashed black line is elastic theory, and dotted curves are the fitted initial response.

Coupon 10

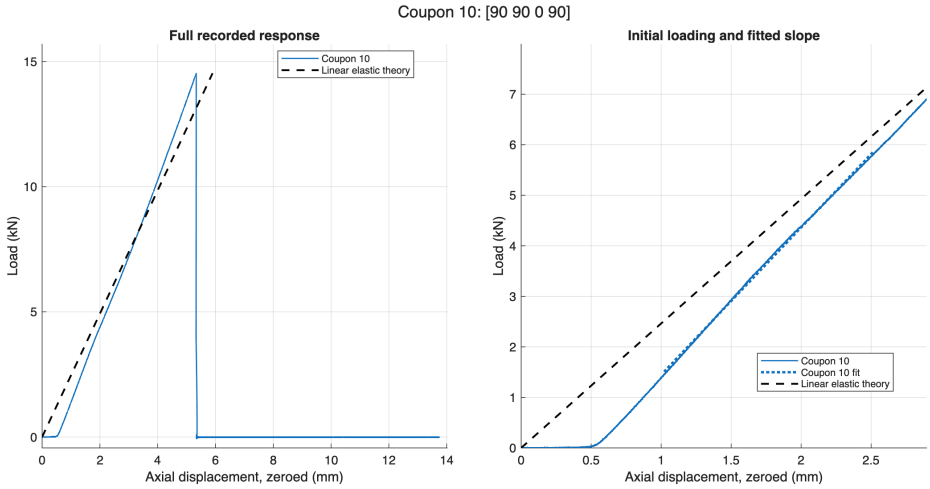


Figure 10. Full recorded response (left) and initial-loading detail (right). Solid curves are measured data, the dashed black line is elastic theory, and dotted curves are the fitted initial response.

Theory uses zero applied moments and free bending, with the listed specimen length as the reference measurement length. Its plotted endpoint is limited by the recorded test range for readability and is not a predicted failure load.